



V93000 Specifications

V93000 SOC Device Power Supplies and VI Cards

Published on 2021-09-29

Notices

Legal Notices

All rights reserved. All text and figures included in this publication are the exclusive property of Advantest Corporation. Reproduction of this publication in any manner without the written permission of Advantest Corporation is prohibited. Information in this document is subject to change without notice.

Trademarks and Registered Trademarks

- ADVANTEST is a trademark of Advantest Corporation.
- All other marks referenced herein are trademarks or registered trademarks of their respective owners.

Warranty

The material contained in this document is provided "as is," and is subject to being changed, without notice, in future editions. Further, to the maximum extent permitted by applicable law, Advantest disclaims all warranties, either express or implied, with regard to this material and any information contained herein, including but not limited to the implied warranties of merchantability and fitness for a particular purpose. Advantest shall not be liable for errors or for incidental or consequential damages in connection with the furnishing, use, or performance of this material or of any information contained herein. Should Advantest and the user have a separate written agreement with warranty terms covering the material in this document that conflict with these terms, the warranty terms in the separate agreement shall control.

Technology Licenses

The hardware and/or software described in this document are furnished under a license and may be used or copied only in accordance with the terms of such license.

Restricted Rights Legend

If software is for use in the performance of a U.S. Government prime contract or subcontract, Software is delivered and licensed as "Commercial computer software" as defined in DFAR 252.227-7014 (June 1995), or as a "commercial item" as defined in FAR 2.101(a) or as "Restricted computer software" as defined in FAR 52.227-19 (June 1987) or any equivalent agency regulation or contract clause. Use, duplication or disclosure of Software is subject to Advantest's standard commercial license terms for 93000 systems and non-DOD Departments and Agencies of the U.S. Government will receive no greater than Restricted Rights as defined in FAR 52.227-19(c)(1-2) (June 1987). U.S. Government users will receive no greater than Limited Rights as defined in FAR 52.227-14 (June 1987) or DFAR 252.227-7015 (b)(2) (November 1995), as applicable in any technical data.

Safety Notices

▲ CAUTION A CAUTION notice denotes a hazard. It calls attention to an operating procedure, practice, or the like that, if not correctly performed or adhered to, could result in damage to the product or loss of important data. Do not proceed beyond a CAUTION notice until the indicated conditions are fully understood and met.

▲ WARNING A WARNING notice denotes a hazard. It calls attention to an operating procedure, practice, or the like that, if not correctly performed or adhered to, could result in death or severe injury. Do not proceed beyond a WARNING notice until the indicated conditions are fully understood and met.

DC Scale Device Power Supplies and VI Cards

Further information

For more information about the Advantest V93000 SOC Series, please visit www.advantest.com.

Contact information

For more information about the Advantest V93000 SOC cards, please contact your local Advantest sales representative.

This information is subject to change without notice.

DC Scale DPS128 Device Power Supply (E8023CS/E8023CSH) - high current specifications

Configuration

	E8023CSH	E8023CS
Maximum number of channels per board	64	128
Maximum source/sink current per channel	1.0 A @ 2.5 V	1.0 A @ 2.5 V
Maximum source/sink current per channel	0.5 A @ 7 V	0.5 A @ 7 V
Maximum source/sink current per board	64 A /64 A@ 2.5 V	66 A /58 A@ 2.5 V
Voltage range	-2.5 V to +7 V	-2.5 V to +7 V

The DPS128/DPS64 card features two power domains with a dedicated power budget. Each domain is able to deliver half of the overall power budget. The power domain to channel mapping must be observed for DPS128 cards when assigning the available channels to the DUT pins. For more information refer to TDC topic DPS128/DPS64 power budget (147811).

	A - test head ^[1]	Compact test head ^[1]	Small test head ^[1]	Large test head ^[1]
Maximum number of boards per system	8	16	32	32
Maximum number of channels per system (E8023CSH)	512	1024	2048	2048
Maximum number of channels per system (E8023CS)	1024	2048	2048	2048

- Parallel connection (ganging) possible for any number of successive channels within a board.
- Up to 64 ganging groups of successive channels are supported within each card.
- Maximum source/sink current per board cannot be exceeded.
- Within a test head, a DPS128 card can be mixed with any other supported device power supply.
- Requires E7010E or E7010F Enhanced Diagnostics Interconnect Board.
- E8023CS only: 32 boards per system only possible if 4 pogo-block version is used (2 channels ganged to 1 channel and 2.0A @2.5V per channel). For more information about the E8023PC pogo cable see TDC topic 141738.

Supply voltage / current range specifications (high current)

Voltage force	Range	Resolution	Accuracy
	-2.5 V to 7 V	200 μ V	± 3 mV

^[1] Applies to test head with DUT Scale interface and standard pogo block

Voltage measure	Range	Resolution	Accuracy
	-2.5 V to 7 V	200 μ V	± 2 mV ^[2] ± 1 mV ^[3]

Current force (clamp)	Range	Resolution	Accuracy
	200 mA to 1.0 A	50 μ A	-3%, +5% of range
	40 mA to 200 mA	10 μ A	-3%, +5% of range
	20 mA to 100 mA	5 μ A	-3%, +5% of range
	5 mA to 25 mA	1 μ A	-3%, +5% of range
	2.5 mA to 12.5 mA	0.5 μ A	-3%, +5% of range
	500 μ A to 2.5 mA	100 nA	-3%, +5% of range
	250 μ A to 1.25 mA	50 nA	-3%, +5% of range
	50 μ A to 250 μ A	10 nA	-3%, +5% of range
	25 μ A to 125 μ A	5 nA	-3%, +5% of range
	5 μ A to 25 μ A	1 nA	-3%, +5% of range
	2.5 μ A to 12.5 μ A	0.5 nA	-3%, +5% of range
Current force (n channels ganged)	n x 200 mA to n x 1.0 A	n x 50 μ A	-3%, +5% of total ganged current

High accuracy current force ^[4]	Range	Resolution	Accuracy
	200 mA to 1.0 A	50 μ A	$\pm$ (3 mA +0.2 % of setting)
	40 mA to 200 mA	10 μ A	$\pm$ (0.6 mA +0.2 % of setting)
	20 mA to 100 mA	5 μ A	$\pm$ (0.6 mA +0.2 % of setting)
	5 mA to 25 mA	1 μ A	$\pm$ (75 μ A +0.2 % of setting)
	2.5 mA to 12.5 mA	0.5 μ A	$\pm$ (75 μ A +0.2 % of setting)
	500 μ A to 2.5 mA	100 nA	$\pm$ (7.5 μ A +0.2 % of setting)
	250 μ A to 1.25 mA	50 nA	$\pm$ (7.5 μ A +0.2 % of setting)
	50 μ A to 250 μ A	10 nA	$\pm$ (750 nA +0.2 % of setting)
	25 μ A to 125 μ A	5 nA	$\pm$ (750 nA +0.2 % of setting)
	5 μ A to 25 μ A	1 nA	$\pm$ (120 nA +0.2 % of setting)
	2.5 μ A to 12.5 μ A	0.5 nA	$\pm$ (120 nA +0.2 % of setting)
Current force (n channels ganged)	n x 200 mA to n x 1.0 A	n x 50 μ A	$\pm$ (n x 3mA +0.2 % of total ganged current)

^[2] Requires 4 averaging samples under pattern control.

^[3] Up to 200 mA forced current, base calibration and auto calibration period 1 month

^[4] Requires the use of a suitable API for high accuracy force current mode and at least 1 # total additional resistive load (incl. DUT) in the Force path

Current measurement

Range	Resolution	Accuracy
± 1.0 A	50 µA	± (1 mA + 0.1 % of reading) ^[5]
± 200 mA	10 µA	± (250 µA + 0.1 % of reading) ^[6]
± 100 mA	5 µA	± (250 µA + 0.1 % of reading) ^[6]
± 25 mA	1 µA	± (25 µA + 0.1 % of reading) ^[7]
± 12.5 mA	0.5 µA	± (25 µA + 0.1 % of reading) ^[7]
± 2.5 mA	100 nA	± (2.5 µA + 0.1 % of reading) ^[8]
± 1.25 mA	50 nA	± (2.5 µA + 0.1 % of reading) ^[9]
± 250 µA	10 nA	± (250 nA + 0.1 % of reading) ^[10]
± 125 µA	5 nA	± (250 nA + 0.1 % of reading) ^[11]
± 25 µA	1 nA	± (50 nA + 0.2 % of reading) ^[11]
± 12.5 µA	0.5 nA	± (50 nA + 0.2 % of reading) ^[11]
n x (±1.0 A) (n channels ganged)	n x 50 µA	± (n x 1 mA + 0.1 % of total current)

Supported "current force" and "current measurement" range combinations

In voltage force mode, current measure range can be changed without prior disconnect.

^[5] Requires 16 averaging samples under pattern control. Maximum supported external capacitance to achieve accuracy is 220 µF. More samples required for larger capacitances.

^[6] Requires 16 averaging samples under pattern control. Maximum supported external capacitance to achieve accuracy is 47 µF. More samples required for larger capacitances.

^[7] Requires 16 averaging samples under pattern control. Maximum supported external capacitance to achieve accuracy is 15 µF. More samples required for larger capacitances.

^[8] Requires 16 averaging samples under pattern control. Maximum supported external capacitance to achieve accuracy is 4.7 µF. More samples required for larger capacitances.

^[9] Requires 16 averaging samples under pattern control. Maximum supported external capacitance to achieve accuracy is 2.2 µF. More samples required for larger capacitances.

^[10] Requires 64 averaging samples under pattern control. Maximum supported external capacitance to achieve accuracy is 4.7 µF. More samples required for larger capacitances.

^[11] Requires 64 averaging samples under pattern control. Maximum supported external capacitance to achieve accuracy is 2.2 µF. More samples required for larger capacitances.

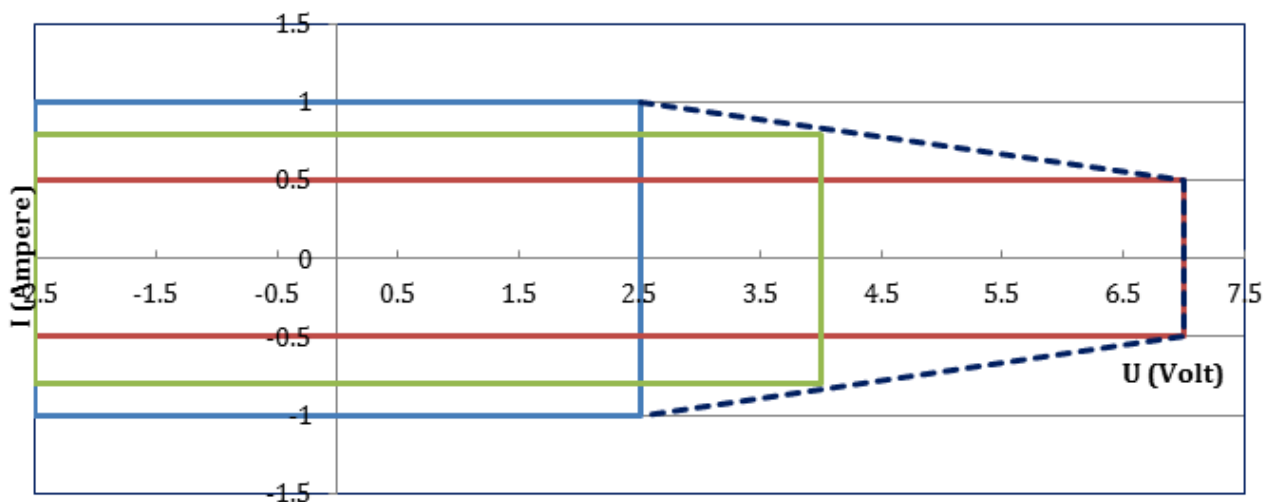
		Current force range										
Current measure range		12.5 μA	25 μA	125 μA	250 μA	1.25 mA	2.5 mA	12.5 mA	25 mA	100 mA	200 mA	1 A
	±12.5 μA	ok		ok		ok		ok		ok		ok
	±25 μA		ok		ok		ok		ok		ok	
	±125 μA			ok		ok		ok		ok		ok
	±250 μA				ok		ok		ok		ok	
	±1.25 mA					ok		ok		ok		ok
	±2.5 mA						ok		ok		ok	
	±12.5 mA							ok		ok		ok
	±25 mA								ok		ok	
	±100 mA									ok		ok
	±200 mA										ok	
	±1 A											ok

DPS128 power diagram high current

The programmed voltage determines the maximum current. If 7 V is programmed, the maximum current is 0.5 A (red box), if 2.5 V is programmed the maximum current is 1 A for all voltages below 2.5 V (blue box).

The power diagram applies to an additional voltage drop in the Force path beyond the pogo pin of ≤ 0.1 V.

Output characteristic: I versus U

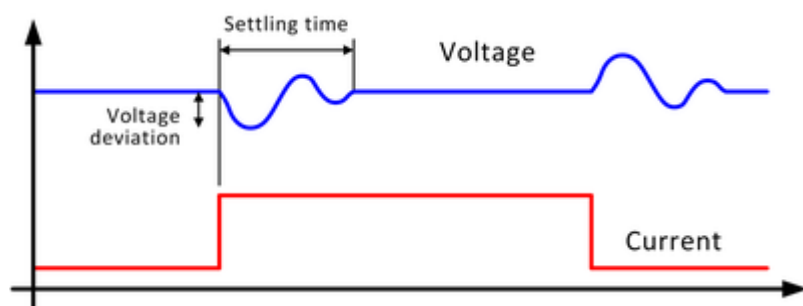


For voltages between 2.5 V and 7 V (green box), the maximum current can be calculated as follows:

$$I_{\text{green}} = \frac{I_{\text{red}} - I_{\text{blue}}}{U_{\text{red}} - U_{\text{blue}}} (U_{\text{green}} - U_{\text{blue}}) + I_{\text{blue}}$$

Dynamic behavior

Explanation of terms: Response time



Settling time after loadstep

at 2 V, settling into ± 20 mV of programmed value

Loadstep	1 μF	47 μF	100 μF
0.1 A to 1.0 A	20 μs	25 μs	30 μs
1.0 A to 0.1 A	20 μs	25 μs	30 μs

Voltage deviation after loadstep

Loadstep	1 μF	47 μF	100 μF
0.1 A to 1.0 A	320 mV	120 mV	85 mV
1.0 A to 0.1 A	500 mV	125 mV	85 mV

Settling wait times defined for DPS128 measurement ranging

The following wait times are to be considered when changing the DPS128 measurement range:

- Range settling time: The required wait time to achieve the specified accuracy of the target range.
- Auto-range execution time: The required wait time to allow the auto-range algorithm to finish.

The following table shows the sum of the range settling time and the auto-range execution time. These wait times are to be considered in the program. Potential device related settling times need to be added on top.

From range	to range					
	1 A	200 mA, 100 mA	25 mA, 12.5 mA	2.5 mA, 1.25 mA	250 μA , 125 μA	25 μA , 12.5 μA
1 A	0 μs	520 μs	790 μs	2020 μs	11310 μs	87320 μs
200 mA, 100 mA	510 μs	0 μs	520 μs	1760 μs	11050 μs	87060 μs
25 mA, 12.5 mA	510 μs	510 μs	0 μs	1340 μs	10640 μs	86650 μs
2.5 mA, 1.25 mA	520 μs	510 μs	510 μs	0 μs	10020 μs	86040 μs
250 μA , 125 μA	520 μs	520 μs	510 μs	1010 μs	0 μs	80020 μs
25 μA , 12.5 μA	520 μs	520 μs	520 μs	1010 μs	8010 μs	0 μs

Legend:

Range down
Range up (Restore range)

Other performance characteristics

Disconnect state	
Input leakage current	< 20 nA (AC relay open)

Voltmeter mode (high impedance mode)	
Voltage range	-2.5 V to +7 V
Voltage resolution	200 μ V
Voltage accuracy	± 1 mV ^[3]
Input leakage current	<30 nA

Compliance range (at disconnect state)	
Maximum external voltage at force input	-6 V to +12 V
Maximum external voltage at sense input	-7 V to +16 V

Load capacitance range	
Required external capacitance for high frequency noise filtering	1nF to 10nF ^[12]
Maximum supported external capacitance with stability maintained	470 μ F per channel (n x 470 μ F up to 5 mF if ganged) ^[13]
Required external capacitance range for DPS style operation	1 - 100 μ F per channel ^[13]

Slew rate control	
Slew rate control	On-, off-, change level- rate individually controlled for each channel

DC profiling	
DC profiling	Supported per channel for voltage and current profiles
Sample rate	Up to 100 ksps

Data analysis	
Data analysis	raw data are stored on board without software interaction and may be uploaded for further computation. Averaging, min and max number are calculated and stored on board
Result analysis	built-in result analysis available that indicate the measurement data is pass or fail against a user specified test limit

Surge tracker

Voltage or current surges are detected against programmable min. and max. values. The surge tracker always checks against both high (positive) and low (negative) thresholds, which can be set individually. Two flags indicate if thresholds have been exceeded. These flags are latched (stored) and can be retrieved any time in the testflow.

^[12] It is recommended to place a low ESL capacitance close to the pogo location. The detailed setup depends on the configuration of the DUT interface board. For more details see topic 149015 in the DUT Board Design Guide.

^[13] In addition to "Required external capacitance for high frequency noise filtering".

Voltage compare mode

Minimum detectable pulse width	1 μ s
Threshold voltage accuracy	± 10 mV typical
Input and threshold range	-2.5 V to +7 V

Current compare mode

Minimum detectable pulse width ^[14]	1 μ s
Threshold current accuracy	± 10 % of selected current range typical
Input and threshold range	Same as current measurement range

Analog performance**AWG mode (16 bit DAC)**

Step rate	200 ksps (kilo samples per second)
Analog bandwidth (-3 dB, 1 V swing)	5 kHz
Total channel memory	28 MB

Digitizer mode (16 bit ADC)

Sample rate	90 ksps (kilo samples per second)
Signal path bandwidth (-3dB)	20 kHz
Total channel memory	28 MB, 11 to 22 bytes required per sample

Trigger functions

Trigger	sequencer#controlled (no external trigger pin required)
---------	---

^[14] in current range 1A

DC Scale UHC4T Device Power Supply (E8024UH)

UHC4T is a device power supply card for the V93000 ATE platform.

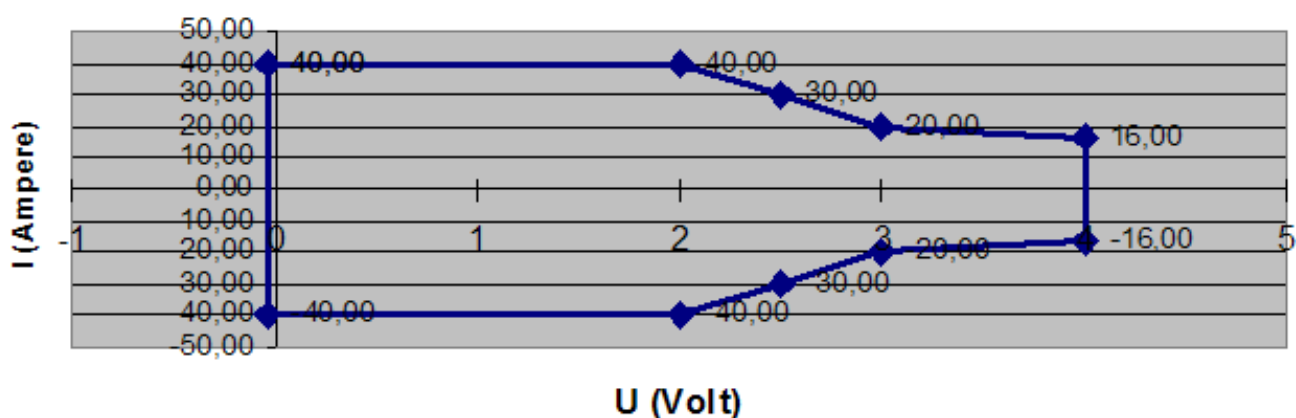
Configuration

Maximum number of channels per board	4
Maximum source/sink current per channel	40 A @ 2.0 V
Maximum source/sink current per channel	30 A @ 2.5 V
Maximum source/sink current per channel	20 A @ 3.0 V
Maximum source/sink current per channel	16 A @ 4.0 V
Maximum source/sink current per board	160 A @ 2.0 V
Maximum source/sink current per system	Maximum number of boards x 160 A
Maximum voltage	0 V to +4 V

Channel and board maxima

	A test head ^[15]	Compact test head ^[15]	Small test head ^[15]	Large test head
Maximum number of boards per system	8	16	32	64
Maximum number of channels per system	32	64	128	256

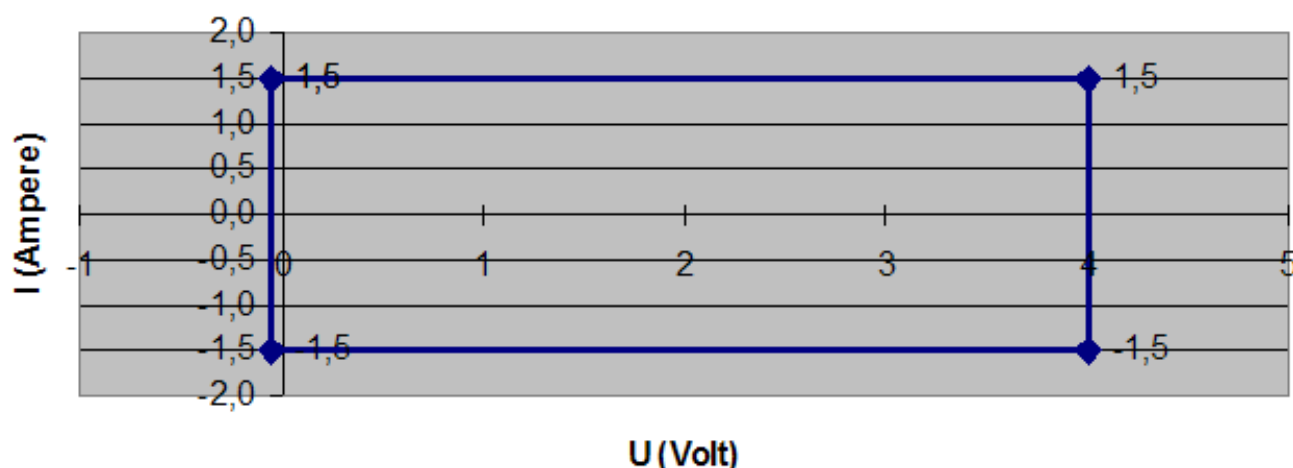
Output characteristic: I versus U
High power mode



UHC4T power diagram high power mode

^[15] Applies to test head with small DUT interface and DUT Scale interface

Output characteristic: I versus U Precision mode



UHC4T power diagram precision mode

- The E8024UH UHC4T requires SmarTest 7.3.2 or higher
- Parallel connection (ganging) possible for any number of successive channels within a card and any number of E8024UH cards within a test head.
- The maximum current per board cannot be exceeded.
- The UHC4T requires a universal DC/DC. An adjacent card must be a UHC4T again (located in a pair of uneven/even card cage slots).
- Static sink power on a board or both boards that are connected to the same DC/DC can be larger than the source power within these limits for each channel of a board:
 - $\leq 0.6 \text{ V} : \leq 40 \text{ A}$
 - $\leq 1.0 \text{ V} : \leq 7 \text{ A}$
 - $\leq 2.5 \text{ V} : \leq 4 \text{ A}$
 - $\leq 4.0 \text{ V} : \leq 2 \text{ A}$
- Dynamic sink power (dis-charge) is possible for supported capacitance.
- Either the high-power mode or the precision mode of a channel is active for force and measure. During the range transition from 1.5 to 10/40 A, both modes are active to avoid glitches.

On the pogo block, four DUT_CAP_ON lines (DUT_CAP_ON_1 through DUT_CAP_ON_4) are available to switch the big blocking capacitance on the DUT board on or off.

If the 40 A range is selected, the signal level is 0 V. If the 1.5 A range is selected, the signal level is 5 V.

- The UHC4T card requires no Enhanced Diagnostics Interconnect Board.

Supply voltage and current range specifications

	Range	Resolution	Accuracy
Voltage force	0 to 4.0 V	200 μV	$\pm 2.5 \text{ mV}$
Voltage measure	0 to 4.0 V	200 μV	$\pm 2 \text{ mV}^{[16]}$
Current force (clamp)	10 to 40 A	2 mA	$\pm(100 \text{ mA} + 0.5\% \text{ of setting})$
	0.5 to 10 A	2 mA	$\pm(50 \text{ mA} + 0.2\% \text{ of setting})$
	15 mA to 1.5 A	100 μA	$\pm 10 \text{ mA}$

^[16] Requires 8 averaging samples under pattern control.

	Range	Resolution	Accuracy
Current force (n channels ganged)	n x 0.5 A to n x 40 A	n x 2 mA	$\pm(n \times 100 \text{ mA} + 0.5\% \text{ of total ganged current})$
Current measurement	$\pm 40 \text{ A}$	2 mA	$\pm(50 \text{ mA} + 0.5\% \text{ of reading})^{[17]}$
	$\pm 10 \text{ A}$	2 mA	$\pm(20 \text{ mA} + 0.2\% \text{ of reading})^{[17]}$
	$\pm 1.5 \text{ A}$	100 μA	$\pm(1.5 \text{ mA} + 0.1 \% \text{ of reading})^{[18]}$
	$\pm 50 \text{ mA}$	2.5 μA	$\pm(50 \mu\text{A} + 0.1 \% \text{ of reading})^{[19]}$
	$\pm 2 \text{ mA}$	100 nA	$\pm(2 \mu\text{A} + 0.1 \% \text{ of reading})^{[20]}$
Current measurement (n channels ganged)	n x 40 A	n x 2 mA	$\pm(n \times 50 \text{ mA} + 0.5\% \text{ of total current})$

Fast unganging is supported, resulting in single-channel accuracy for Idd-measurements. This includes DUT capacitance switching using the four DUT_CAP_ON lines.

Supported "Current force" and "Current measurement" range combinations

		Current force range, 10/40 A high-power mode	Current force range, 1.5 A precision mode
Current measurement range, high-power mode	10/40A	yes	no
Current measurement range, precision mode	1.5 A	no	yes
	50 mA	no	yes
	2 mA	no	yes

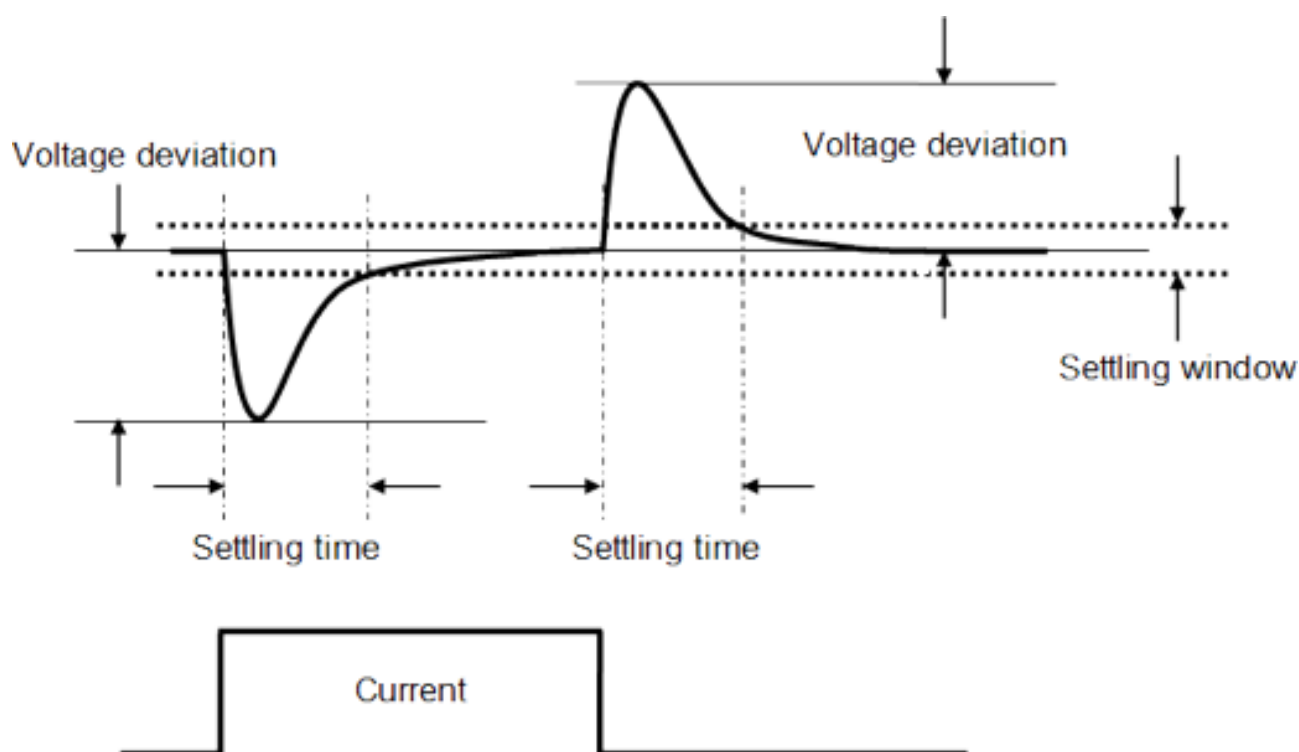
Explanation of terms

^[17] Requires 64 averaging samples under pattern control.

^[18] Requires 256 averaging samples under pattern control.

^[19] Requires 256 averaging samples under pattern control. Maximum supported external capacitance to achieve accuracy is 100 μF .

^[20] Requires 256 averaging samples under pattern control. Maximum supported external capacitance to achieve accuracy is 10 μF .



Explanation of terms

Settling time after load step ($dl/dt = 2.5 \text{ A}/\mu\text{s}$)

At 2 V, settling into $\pm 20 \text{ mV}$ of programmed value, single-channel operation:

Load step	270 μF (bulk)	540 μF (bulk)
10 A to 30 A	0 μs	0 μs
30 A to 10 A	0 μs	0 μs

At 2 V, settling into $\pm 10 \text{ mV}$ of programmed value, single-channel operation:

Load step	270 μF (bulk)	540 μF (bulk)
10 A to 30 A	15 μs	15 μs
30 A to 10 A	18 μs	18 μs

Settling time after load step ($dl/dt = 40 \text{ A}/\mu\text{s}$)

At 2 V, settling into $\pm 20 \text{ mV}$ of programmed value, single-channel operation:

Load step	1080 μF (bulk), 100 μF (ceramic)	2160 μF (bulk), 200 μF (ceramic)
10 A to 30 A	7 μs	7 μs
30 A to 10 A	7 μs	6 μs

At 2 V, settling into $\pm 10 \text{ mV}$ of programmed value, single-channel operation:

Load step	1080 μF (bulk), 100 μF (ceramic)	2160 μF (bulk), 200 μF (ceramic)
10 A to 30 A	9 μs	8 μs

Load step	1080 μ F (bulk), 100 μ F (ceramic)	2160 μ F (bulk), 200 μ F (ceramic)
30 A to 10 A	10 μ s	7 μ s

Settling time after load step ($dI/dt = 400$ A/ μ s)

At 2 V, settling into ± 20 mV of programmed value, ganging four channels on one board:

Load step	4 mF (bulk), 200 μ F (ceramic)	8 mF (bulk), 400 μ F (ceramic)
40 A to 120 A	4 μ s	5 μ s
120 A to 40 A	9 μ s	4 μ s

At 2 V, settling into ± 10 mV of programmed value, ganging four channels on one board:

Load step	4 mF (bulk), 400 μ F (ceramic)	8 mF (bulk), 400 μ F (ceramic)
40 A to 120 A	10 μ s	7 μ s
120 A to 40 A	13 μ s	10 μ s

Voltage deviation after load step ($dI/dt = 2.5$ A/ μ s)

At 2 V, single-channel operation:

Load step	270 μ F (bulk)	540 μ F (bulk)
10 A to 30 A	18 mV	18 mV
30 A to 10 A	18 mV	18 mV

Voltage deviation after load step ($dI/dt = 40$ A/ μ s)

At 2 V, single-channel operation:

Load step	1080 μ F (bulk), 100 μ F (ceramic)	2160 μ F (bulk), 200 μ F (ceramic)
10 A to 30 A	53 mV	35 mV
30 A to 10 A	55 mV	35 mV

Voltage deviation after load step ($dI/dt = 400$ A/ μ s)

At 2 V, ganging four channels on one board:

Load step	4 mF (bulk), 200 μ F (ceramic)	8 mF (bulk), 400 μ F (ceramic)
40 A to 120 A	50 mV	35 mV
120 A to 40 A	80 mV	60 mV

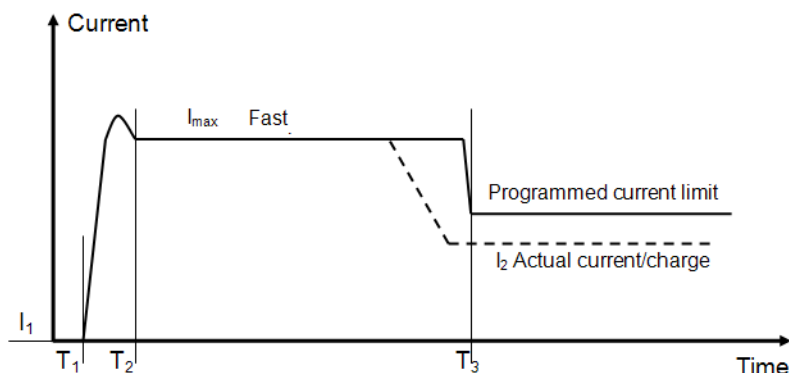
Modes and warnings**Fast precharge**

Maximum supported current	40 A (independent of actual voltage setting) is forced for a programmable time (0-10 ms), independent of actual current limit settings, if the 10/40 A current force range is selected
---------------------------	--

Fast precharge

Programmable timing resolution	1 μ s
Precharge start	When DUT is connected
Mode selection	Explicitly enabled or disabled with a firmware command (default condition is disabled) or implicitly with a programmed time in an equation

- The fast precharge settings are identical for all channels of a gang group
- This mode is not enabled for voltage changes



At T_1 current starts to flow after connect was initiated (HIZ or LOZ connect)
 $T_2 - T_1$ is settling time
 At T_2 I_{max} flows for fast precharge
 $T_3 - T_1$ is fast precharge time (programmable with 1 μ s resolution between 0 - 10 ms)
 At T_3 current is reduced to I_2 which is equal to or smaller than programmed current limit. If the programmed fast precharge time is too long, current drops to actual load before T_3 is reached

Fast precharge mode**Peak mode**

Peak current	Peak current > 40 A for a few microseconds
Mode selection	Enabled, no control flag or alarm is set

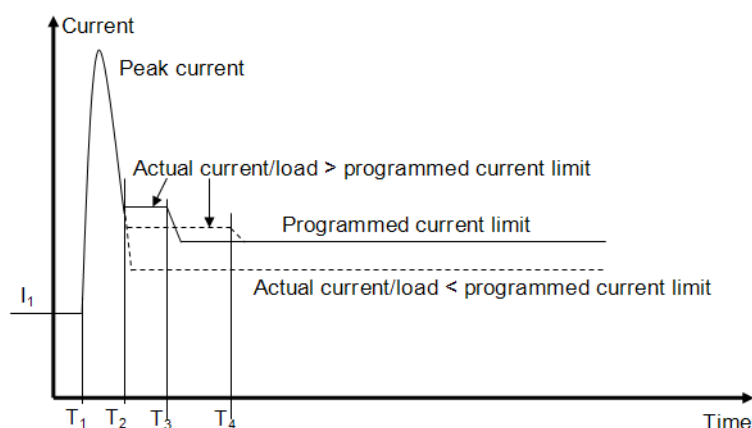
Current limit

Typical times to limit the current at a current limit of 10 A (one channel, no ganging, 1080 μ F)

Load step	Response time
0 A to 40 A	600 μ s
8 A to 24 A	1.3 ms
8 A to 16 A	3.2 ms

- Activation time of current limit depends on the actual load (overdrive of comparator)

Load change (constant current mode selected)



At T_1 load change occurs
 Peak current is delivered for few microseconds
 At T_2 current is reduced to actual current /load, over current flag is triggered. Flag is reset, if actual current < programmed current limit
 $T_{3,4} - T_2$ is current limit activation time which depends on overdrive of limit comparator
 At $T_{3,4}$ current limit becomes active and current is limited to programmed current limit

Constant current mode

Constant current mode

Constant current mode	Maximum allowed current is equal to actual programmed current limit
-----------------------	---

Disconnect mode

Disconnect mode	If current exceeds the set current limit, the channel is disconnected
-----------------	---

Disconnect delay time	Programmable (0 - 65 ms, timing resolution 1 μ s)
-----------------------	---

Mode selection	Explicitly enabled or disabled with a firmware command (default condition is disabled) or implicitly with a programmed time in an equation
----------------	--

- Time delay counter is triggered by the overcurrent flag.
- The settings are identical for all channels of a gang group.
- This mode is selectable in all current force ranges.
- The safety line is not activated (same behavior as HC-DPS).

Overvoltage protection

Allowed voltage drop	Individually programmable for force and return path between 0 - 500 mV, programming resolution is 1 mV
----------------------	--

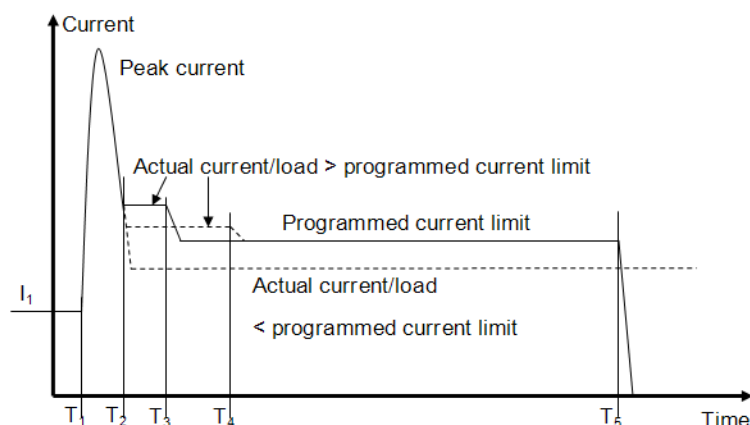
Total allowed voltage drop	500 mV
----------------------------	--------

Mode selection	Always on, in all current ranges, except for the time the fast precharge mode is on
----------------	---

Default voltage limit	250 mV for both the force and return path
-----------------------	---

- Force voltage at output of UHC4T is continuously compared to voltage at charge. If the difference is above the programmed limit, the output channel is disconnected. The voltage drop can be measured with the UHC4T.
- There are separate comparators for both the force and the return line on both the source and the sink path (four in total).
- The safety line is not activated in case of alarm (same behavior as HC-DPS).

Load change (disconnect mode selected)



At T₁ load change occurs, peak current is delivered for few microseconds
 At T₂ current is reduced to actual current/load
 T_{3,4} - T₂ is current limit activation time which depends on overdrive of limit comparator
 At T_{3,4} current limit becomes active and current is limited to programmed current limit
 Over current flag is triggered, if actual current/load > programmed current limit
 Flag is reset, if actual current < programmed current limit
 At T₅ the channel is disconnected (HI-Z or LO-Z)
 T₅ - T_{3,4} is disconnect time (programmable with 1 μs resolution between 0 - 65 ms)

Disconnect mode

Other protection mechanisms (one line per board)

Safety line	Connected to DUT board via pogo bock. Internally pulled high to +5V from DUT board with a resistor of 46.4k. When pulled low, all channels of UHC are disconnected. Can be used to simultaneously disconnect all UHC4T channels by an external alarm switch.
+5 V from DUT board	If not applied, all UHC4T channels are disconnected. Can be used to detect improperly connected DUT board.

Other performance characteristics

Disconnect state

Input leakage current	< 100 nA @ ≤ 3.4 V (AC relay open)
-----------------------	------------------------------------

Voltmeter mode (high impedance mode)

Voltage range	0 V to +4 V
Voltage resolution	200 μV
Voltage accuracy	±2 mV
Input leakage current	< 100 nA @ ≤ 3.4V (device connected to force and sense lines)

Differential mode (using force sense and ground sense)

Voltage range	Same as voltage measure
Voltage resolution	Same as voltage measure
Voltage accuracy	Same as voltage measure

High speed differential sampling mode (14 bit)

Number of differential inputs	1 per channel, individually addressable and usable within a gang chain
Sample rate	10 Msps (megasamples per second)
Voltage window	-2.5 V to +4 V
Differential voltage range	0 V to 3 V
Voltage resolution	400 μ V
DC accuracy	2 mV
Signal path bandwidth (-3 dB)	5 MHz
Total channel memory	224 MB

AWG mode (16 bit DAC)

Step rate	1 Msps (megasamples per second)
Analog bandwidth (-3 dB, 3 V swing)	3 kHz
Total channel memory	224 MB
Vbump function	Sequencer#controlled, programmable

Digitizer mode (16 bit ADC)

Sample rate	500 ksps (kilosamples per second)
Signal path bandwidth (-3 dB)	25 kHz
Total channel memory	224 MB

Trigger functions

Trigger	Sequencer#controlled (no external trigger pin required)
---------	---

Data analysis

Data analysis	Raw data is stored on board without software interaction and may be uploaded for further computation. Averaging, min and max number are calculated and stored on board.
---------------	---

Data analysis

Result analysis	Built-in result analysis available that indicates the measurement data is pass or fail against a user specified test limit.
-----------------	---

Compliance range (at disconnect state)

Maximum external voltage at force input	-3 V to +10 V
Maximum external voltage at sense input	-3 V to +10 V
Maximum external voltage at sampling input	-3 V to +10 V
Maximum external voltage at safety line	-3 V to +5.5 V
Maximum external voltage at +5 V DUT signal	-3 V to +5.5 V

Settling time after re-programming (settling within given band of programmed step or within accuracy band, 2 x 270 μ F capacitive charge)

Voltage force (0-2 V step)	200 μ s ($\pm$ 1% band)
Current force / measure (0-10 A step)	40 μ s (+0 -20% band) 4 ms ($\pm$ 1% band)

Charge capacitance range

Minimum required external capacitance	270 μ F per channel for high power mode 1 μ F per channel for precision mode
Maximum supported external capacitance with stability maintained	10 mF per channel
Recommended external capacitance	1 - 2 mF, see also the explanation for the voltage/current specifications, above

Blocking caps (examples, both types are required on DUT board)

Tantalum (example for bulk capacitance)

Kemet	T530D477M2R5A(1)E010 - 470 μ F/2.5V
Panasonic	OSCON 4SVPC560MX - 560 μ F/4V

Ceramic (example)

TDK	C1608X5R1E105K (1 μ F/25V/0603/X5R-Ceramic), array of 25 each
TDK	C1608X7R1H104K (100nF/50V/0603/X7R-Ceramic), array of 25 each

Power ramp at probe	
Probe card protection	User-programmable sequence for both shorts and open-check, using the precision mode
Thermal control	
Current monitor	Current monitor output per channel provides voltage proportional to the output current, 1 V per 10 A current, reference potential is GSn- line (Ground Sense). NoteSense-lines are sensitive to noise. It it is recommended to use the GND-plane close to the connection of the GSn-line.
Typical accuracy	±5 mV ± 1% of reading

DC Scale XPS256 Universal DPS/VI Card



XPS256 overview

General specifications	
Maximum channel count	256 channels per card
Maximum source current per channel	1.0 A @ -2.5 V ... +3.0 V
	0.6 A @ -2.5 V ... +7.0 V
Maximum sink current per channel	1.0 A @ -2.0 V ... +3.0 V
	0.6 A @ -2.5 V ... +7.0 V
Maximum source current per card	256 A @ -2.5 V ... +2.0 V ^[23]
Maximum sink current per card	230 A @ -2.1 V ... +4.0 V ^[23]
Voltage range	-2.5 V ... +7.0 V
Calibration, operation, environmental	
Warm-up time	60 minutes
Basic maintenance period	6 months
Base calibration period (traceable calibration)	6 months

Calibration, operation, environmental	
Calibration period (system auto adjustment) ^[21]	3 months
Operating temperature range	15 °C to 30 °C (59 °F to 86 °F)
Temperature range for guaranteed specifications	20 °C to 30 °C (68 °F to 86 °F)
Maximum humidity at 30 °C	< 70 % R. H., non-condensing
Storage temperature range (without water)	-40 °C to +70 °C

System configuration	CX	SX	LX
Maximum number of cards per system	9	18	27
Maximum number of channels per system	2304 ^[22]	4608 ^[24]	6912 ^[24]

Other specifications
- Maximum source/sink current per card cannot be exceeded. - Requires E7010F, Rev. 11 or higher or E7010G Enhanced Diagnostics Interconnect Board.

XPS256 voltage specifications

Voltage force	Range	Resolution	Accuracy ^[25] ^[28]
	-0.5 V ... +1.5 V	10 μ V	$\pm$ (150 μ V + 0.02 % of setting)
	-2.5 V ... +7.0 V	50 μ V	$\pm$ (400 μ V + 0.02 % of setting)

Voltage measurement	Range	Resolution	Accuracy ^[26] ^[27]
	-0.5 V ... +1.5 V	10 μ V	$\pm$ (150 μ V + 0.02 % of value)
	-2.5 V ... +7.0 V	50 μ V	$\pm$ (400 μ V + 0.02 % of value)

Voltage clamp	Range	Resolution	Accuracy ^[28]
	-0.5 V ... +1.5 V	10 μ V	$\pm$ (500 μ V + 0.1 % of setting)
	-2.5 V ... +7.0 V	50 μ V	$\pm$ (3 mV + 0.1 % of setting)

^[21] Valid at ambient temperature within ± 2.5 K of calibration temperature.

^[22] Max. 2048 channels with DUT Scale interface and EK751HDA pogo cable assembly

^[23] Combined voltage drop on force and GND line not exceeding ± 0.2 V

^[24] Max. 4096 channels with DUT Scale interface and EK751VHDA pogo cable assembly (max. 2048 channels with EK751HDA)

^[25] Max cap load < 1 mF / channel

^[26] Measurement conditions VBw = 30 kHz, IBw = 10 kHz, measure bandwidth = 30 kHz

^[27] Requires 40 samples (20 μ s).

XPS256 current specifications

Current force	Range	Resolution	Accuracy ^{[31] [32]}
	± 1.0 A	10 µA	± (1 mA + 0.15 % of setting + 1 nA/V * V _{actl})
	± 100 mA	1 µA	± (100 µA + 0.15 % of setting + 1 nA/V * V _{actl})
	± 10 mA	100 nA	± (10 µA + 0.15 % of setting + 1 nA/V * V _{actl})
	± 1 mA	10 nA	± (1 µA + 0.15 % of setting + 1 nA/V * V _{actl})
	± 100 µA	1 nA	± (250 nA + 0.15 % of setting + 1 nA/V * V _{actl})
	± 10 µA	100 pA	± (50 nA + 0.15% of setting + 1 nA/V * V _{actl})
Current force (n channels ganged)	± n * 1.0 A	n * 10 µA	± (n * 1 mA + 0.15 % of setting + n * 1 nA/V * V _{actl})

Current measurement	Range	Resolution	Accuracy ^{[29] [33] [32]}
	± 1.0 A	10 µA	± (1 mA + 0.1 % of value + 1 nA/V * V _{actl})
	± 100 mA	1 µA	± (100 µA + 0.1 % of value + 1 nA/V * V _{actl})
	± 10 mA	100 nA	± (10 µA + 0.1 % of value + 1 nA/V * V _{actl})
	± 1 mA	10 nA	± (1 µA + 0.1 % of value + 1 nA/V * V _{actl})
	± 100 µA	1 nA	± (250 nA + 0.1 % of value + 1 nA/V * V _{actl}) ^[34]
	± 10 µA	100 pA	± (50 nA + 0.1 % of value + 1 nA/V * V _{actl}) ^[34]
Current measure (n channels ganged)	± n * 1.0 A	n * 10 µA	± (n * 1 mA + 0.1 % of value + n * 1 nA/V * V _{actl})

Current clamp	Range	Resolution	Accuracy ^[31]
	10 mA ... 1.0 A	10 µA	± (2 mA + 0.2 % of setting)
	1 mA ... 100 mA	1 µA	± (200 µA + 0.2 % of setting)
	100 µA ... 10 mA	100 nA	± (20 µA + 0.2 % of setting)
	10 µA ... 1 mA	10 nA	± (2 µA + 0.2 % of setting)
	1 µA ... 100 µA	1 nA	± (400 nA + 0.2 % of setting)
	100 nA ... 10 µA	100 pA	± (70 nA + 0.2 % of setting)
Current clamp (n channels ganged)	n * (10 mA ... 1.0 A)	n * 10 µA	± (n * 2 mA + 0.2 % of setting)
Clamp response time (typical)	< 35 µs ^[30]		

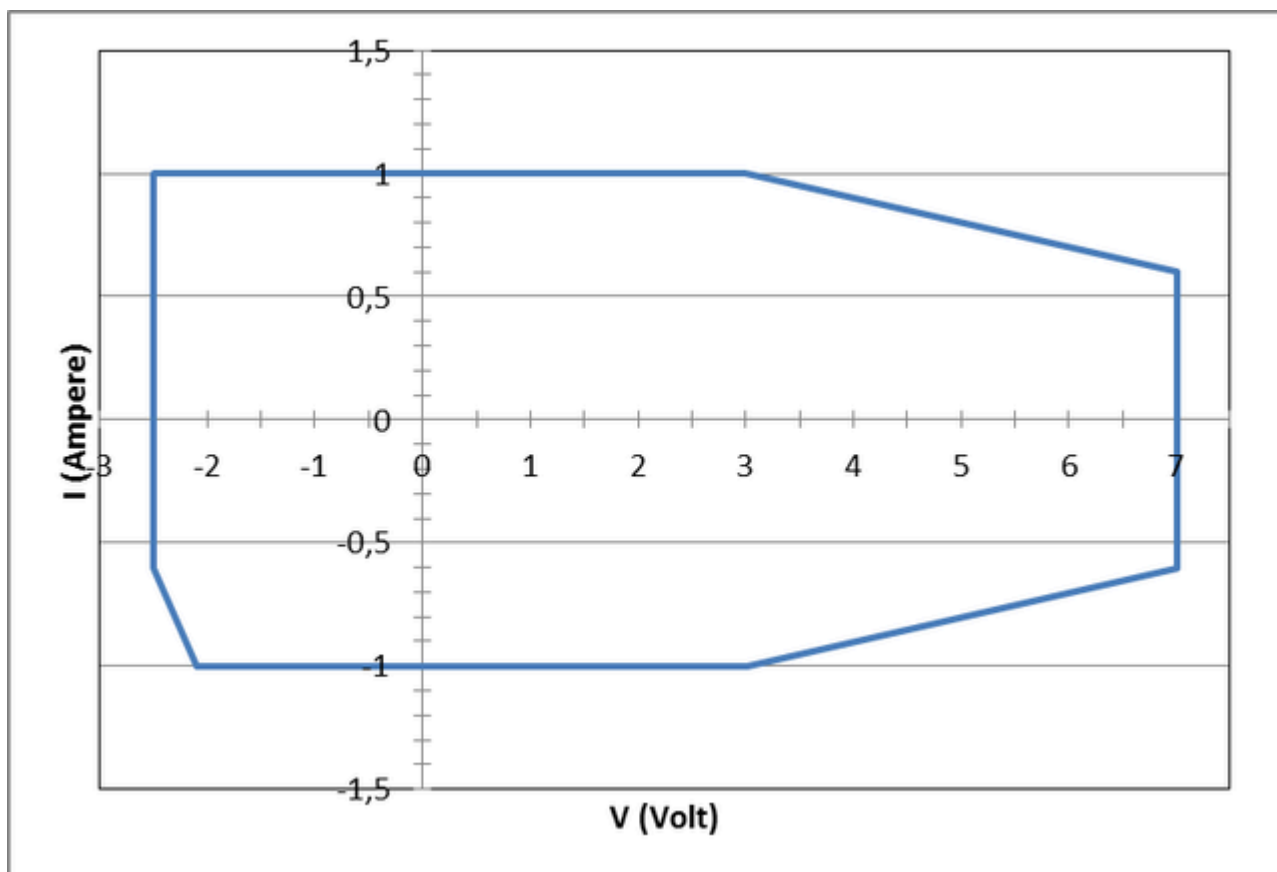
^[28] Measurement conditions VBw = 30 kHz, IBw = 10 kHz

^[29] Measurement conditions VBw = 30 kHz, IBw = 10 kHz, measure bandwidth = 30 kHz

^[30] At Ibandwidth = 50 kHz, into programmed value ± 10 % of Irange, loadstep mode standard

Hardware current limiter

Short circuit current < 1.8 A

XPS256 voltage and current capability specifications*Continuous voltage and current capabilities per channel*

The power diagram applies to an additional voltage drop in the force path beyond the pogo pin of total $\leq 0.5V$.

Maximum continuous voltage and current capability per card

The following diagrams show the capability per card for a different number of active channels and different maximum total voltage drop on the force lines P and GND from the pogo pins to the DUT.

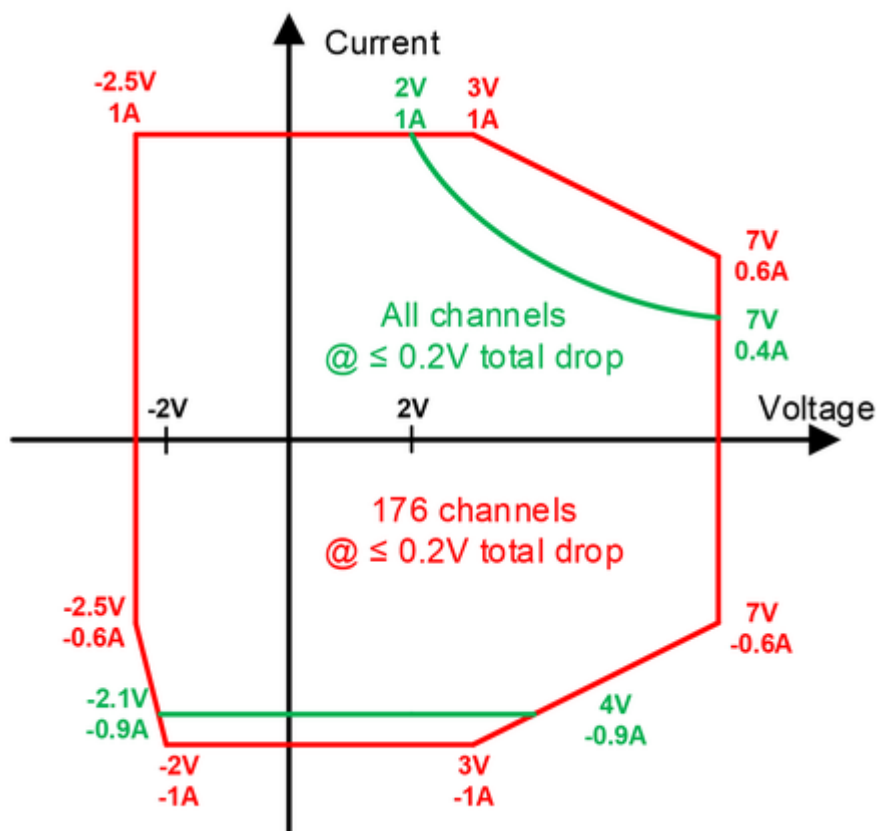
All channels / 176 channels @ 0.2 V maximum total voltage drop for force and return path

[31] Measurement conditions VBw= 30 kHz, IBw= 10 kHz

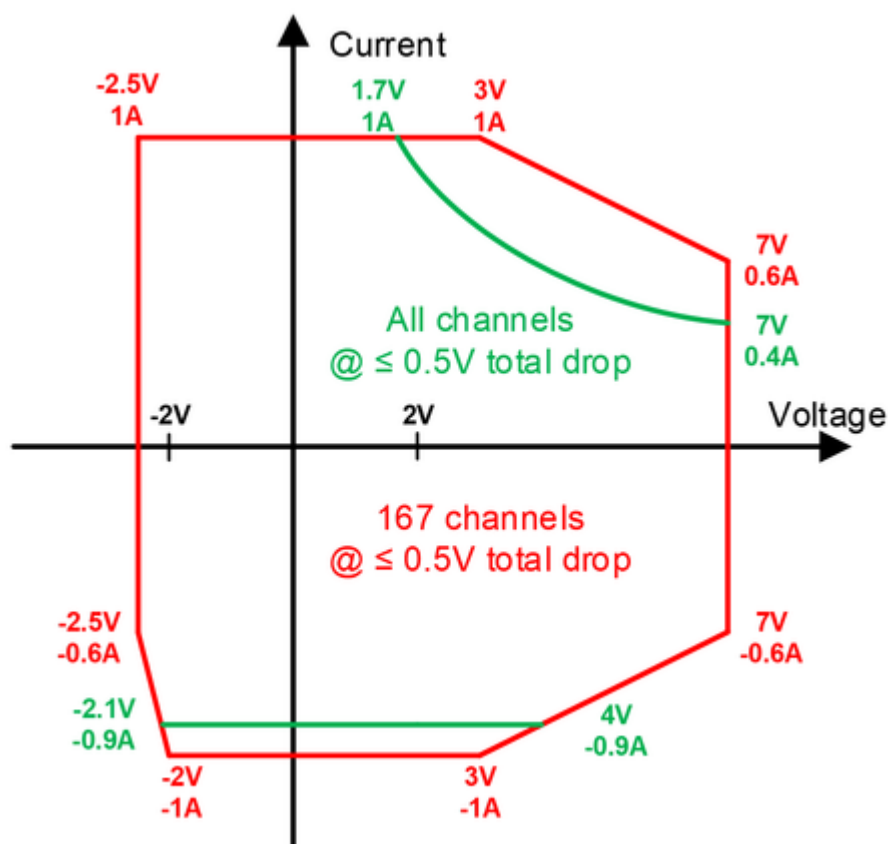
[32] V_{act} is the actual voltage of the channel, which is either the programmed force voltage or the voltage forced by the DUT, depending on the mode the channel is in (forcing or clamping).

[33] Requires 160 samples (80 μs).

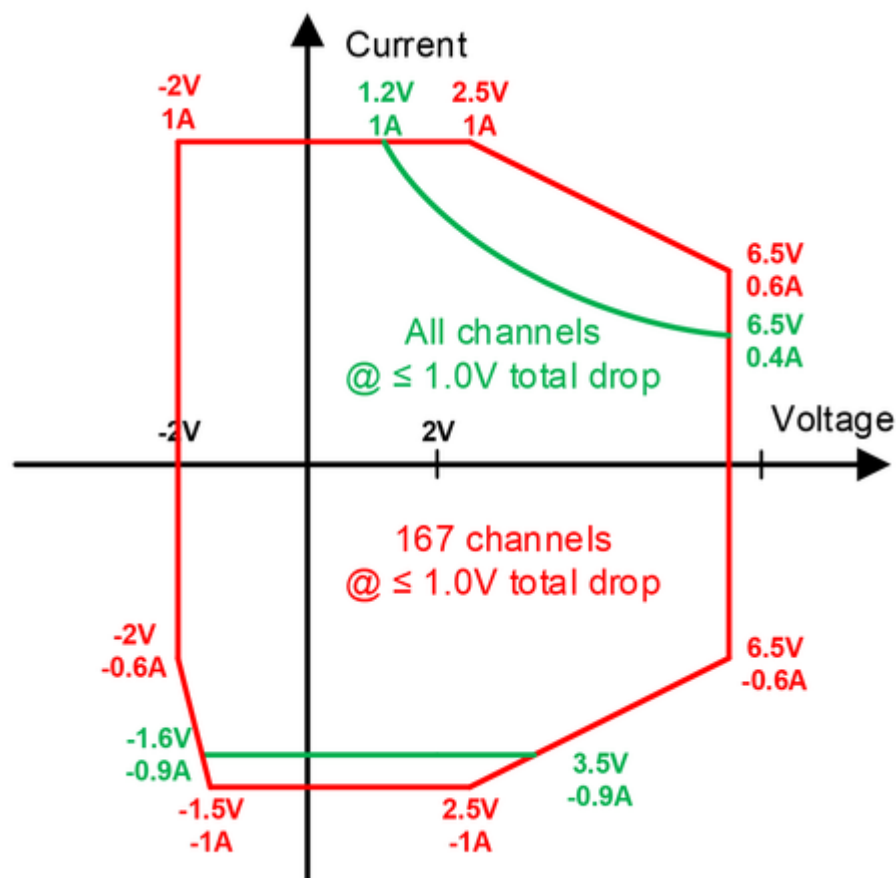
[34] Requires 640 samples (320 μs).



All channels / 167 channels @ 0.5 V maximum total voltage drop for force and return path

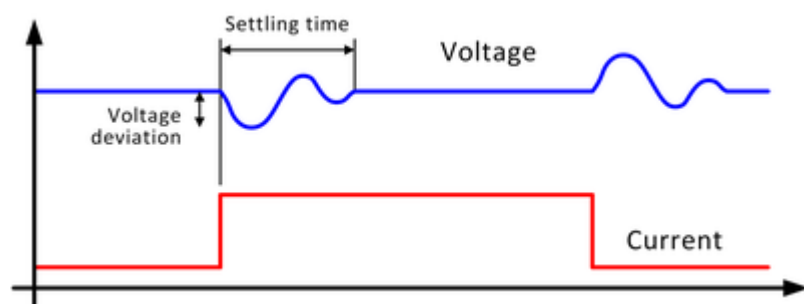


All channels / 167 channels @ 1.0 V maximum total voltage drop for force and return path



XPS256 load step response

Explanation of terms:



Load step response ^[35]	Regulation mode	Load step	100 μF ^[39]	47 μF ^[39]	10 μF ^[36]
Settling time ^[37]	Fast	50 % ^[40]	20 μs	15 μs	10 μs
		80 % ^[41]	20 μs	15 μs	10 μs
	Standard	50 % ^[40]	50 μs	30 μs	20 μs

^[35] Typical at pogo pin, in $n * \pm 1.0 A$ range, clamp setting $n * 1 A$ (n = number of channels ganged)

^[36] Voltage bandwidth: 30 kHz

^[37] At 1 V, settling into $\pm 10 mV$ of programmed value

Load step response ^[35]	Regulation mode	Load step	100 μ F ^[39]	47 μ F ^[39]	10 μ F ^[36]
Maximum voltage deviation ^[38]	Fast	80 % ^[41]	55 μ s	35 μ s	25 μ s
		50 % ^[40]	30 mV	40 mV	85 mV
		80 % ^[41]	55 mV	70 mV	120 mV
	Standard	50 % ^[40]	60 mV	65 mV	110 mV
		80 % ^[41]	90 mV	110 mV	140 mV

XPS256 AWG, digitizer, and comparator specifications

Source waveform mode (AWG)

Regulated mode:

Max. step rate	2 Msps (million samples per second) ^[45]
Analog bandwidth (-3 dB, 1 V swing)	30 kHz (typ.)
Maximum number of samples	14 Msamples (million samples) ^[42] ^[43]
Accuracy	see Voltage force / Current force specifications

Measure waveform mode (Digitizer)

Voltage sampling:

Sample rate	2 Msps (million samples per second) ^[45]
Signal path bandwidth (-3 dB)	500 kHz (typ.)
Maximum number of samples	22 Msamples (million samples) ^[47]
Accuracy	see voltage measure specifications ^[44]

Current sampling:

Sample rate	2 Msps (million samples per second) ^[45]
Signal path bandwidth (-3 dB)	500 kHz (typ.)
Maximum number of samples	22 Msamples (million samples) ^[47]
Accuracy	see current measure specifications ^[44]

^[35] Typical at pogo pin, in $n * \pm 1.0$ A range, clamp setting $n * 1$ A (n = number of channels ganged)

^[36] Voltage bandwidth: 30 kHz

^[38] At 1 V, any number of channels ganged

^[39] Voltage bandwidth: 50 kHz

^[40] Verified at 25 % to 75 % load step

^[41] Verified at 10 % to 90 % load step

^[42] Requires SmarTest 8, 1 Msamples max. with SmarTest 7

^[43] 64 bits per sample (256 bits for SmarTest 7) in total channel memory (896 Mbit per channel, will be shared with other DC action and ganged channels)

Voltage compare mode ^[46]	
Minimum detectable pulse width	1 μ s (typ.)
Threshold voltage accuracy	Same as voltage measure (1 kHz filter)
Input and threshold range	Same as voltage measure

Current compare mode ^[46]	
Minimum detectable pulse width	1 μ s (typ.)
Threshold current accuracy	Same as current measure (1 kHz filter)
Input and threshold range	Same as current measure

XPS256 other performance characteristics

Load capacitance range	
Minimum required external capacitance with stability maintained	0 nF ^[48]
Maximum supported external capacitance with stability maintained	1000 μ F per channel ^[48] (n * 1000 μ F if ganged)

Output noise	
Output noise (spurious up to 20 MHz, typical) ^[49]	100 μ V rms ^[50] 600 μ V rms ^[51]

Slew rate control (Voltage)	
On-, off-, change level-rate individually controlled for each channel	
	1.5 V range 7.0 V range

^[44] With 20 kHz digital filter

^[45] Requires SmarTest 8, 1.5 Msps max. with SmarTest 7

^[46] Requires SmarTest 8

^[47] 40 bits per sample (max. 11 Msamples, 80 bits per sample for SmarTest 7) in total channel memory (896 Mbit per channel), channel memory will be shared with other DC actions (max 16M per action)

^[48] In standard regulation mode, Vbandwidth = 10 kHz

^[49] For single channel, in 1A range

^[50] At 100 μ F per channel capacitive load

^[51] At 0 nF capacitive load

Slew rate control (Voltage)		
Minimum programmable slew rate	10 $\mu\text{V} / \mu\text{s}$	45 $\mu\text{V} / \mu\text{s}$
Maximum programmable slew rate	2 V / μs	2 V / μs

Slew rate control (Current)			
Change level-rate individually controlled for each channel ^[52]			
	Range	Min	Max
	$\pm 1.0 \text{ A}$	10.0 $\mu\text{A}/\mu\text{s}$	200.0 $\text{mA}/\mu\text{s}$
	$\pm 100 \text{ mA}$	1.0 $\mu\text{A}/\mu\text{s}$	20.0 $\text{mA}/\mu\text{s}$
	$\pm 10 \text{ mA}$	100.0 $\text{nA}/\mu\text{s}$	2.0 $\text{mA}/\mu\text{s}$
	$\pm 1 \text{ mA}$	10.0 $\text{nA}/\mu\text{s}$	200.0 $\mu\text{A}/\mu\text{s}$
	$\pm 100 \mu\text{A}$	1.0 $\text{nA}/\mu\text{s}$	20.0 $\mu\text{A}/\mu\text{s}$
	$\pm 10 \mu\text{A}$	100.0 $\text{pA}/\mu\text{s}$	2.0 $\mu\text{A}/\mu\text{s}$

Disconnect mode		
Disconnect mode	If current reaches set current limit, the channel is disconnected	
Disconnect delay time ^[53]	Range	Resolution
	0 μs ... 127 μs	1 μs
	> 127 μs ... 1.27 ms	10 μs
	> 1.27 ms ... 12.7 ms	100 μs
	> 12.7 ms ... 127 ms	1 ms
	> 127 ms ... 1.27 s	10 ms

Disconnect state	
Max. input leakage current (AC relay open, typical)	$\pm (10 \text{ nA} + 1 \text{ nA} / \text{V})$ force and sense connected $\pm (1 \text{ nA} + 1 \text{ nA} / \text{V})$ only sense connected
Voltmeter mode (high impedance mode)	
Voltage range	Same as voltage measure
Voltage resolution	Same as voltage measure
Voltage accuracy	Same as voltage measure
Max. input leakage current (typical)	$\pm (1 \text{ nA} + 1 \text{ nA} / \text{V})$
Max. input voltage (sense / ground sense)	-0.5 V ... +1.5 V in 1.5 V range
	-2.5 V ... +7.0 V in 7.0 V range
Max. differential voltage (sense to ground sense)	-0.5 V ... +1.5 V in 1.5 V range
	-2.5 V ... +7.0 V in 7.0 V range

^[52] Min/max values valid for single channel, for $n \times$ ganging, numbers must be multiplied by n

^[53] Time delay counter is triggered by the overcurrent flag. The settings are identical for all channels of a gang group.

Compliance range (at disconnect state)

Min/max external voltage at force input	-4.5 V ... +7.5 V
Min/max external voltage at sense input	-4.5 V ... +7.5 V

Overvoltage protection

Maximum voltage range	-80 V ... +80 V
Maximum energy of voltage spike	20 mJ
Maximum current	1.0 A

Voltage drop protection / measurement

Allowed voltage drop	Individually programmable for force and return path between 0 and 500 mV, programming resolution is 1 mV Two level sets can be programmed
Total allowed voltage drop	1000 mV
Voltage drop measurement accuracy (relative)	$\pm 4 \text{ mV}$ ^[54]
Voltage drop measurement accuracy (absolute)	$\pm (20 \text{ mV} + 50 \text{ mV} / \text{A})$
Voltage drop measurement resolution	900 μV

DC profiling

DC profiling	Supported per channel for simultaneous voltage and current profiling ^[55]
Maximum sample rate	2 Msps (million samples per second) ^[56]
Maximum number of samples	22 Msamples (million samples)
Input signal bandwidth	500 kHz (typ.)

Surge tracker

Trigger	Voltage or current surges are detected against programmable min. and max. values. The surge tracker always checks against both high (positive) and low (negative) thresholds, which can be set individually. Two flags indicate if thresholds have been exceeded. These flags are latched (stored) and can be retrieved any time in the testflow.
Input signal bandwidth	500 kHz (typ.)
Threshold voltage accuracy (typical)	$\pm 10 \text{ mV}$
Threshold current accuracy (typical)	$\pm 10 \%$ of selected current range

^[54] Delta measurements on same channel at same operating conditions (i.e. force current, same auto-calibration). Requires 40 averages.

^[55] Simultaneous profiling requires SmarTest 8.4.2 or higher.

^[56] Requires SmarTest 8.4.2 or higher, 1 Msps max. with SmarTest 7, valid for profiling one parameter only

XPS256 pogo block specifications

EK751HDA - High Density Pogo Cable for EK751 (16 x 16 channels)

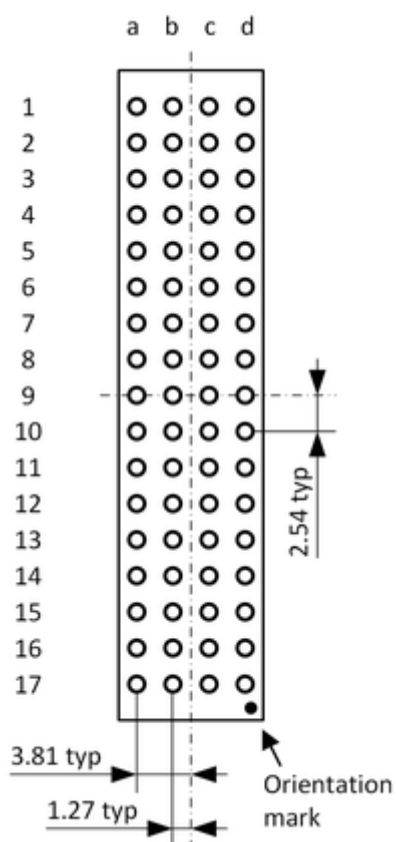
High-density (HD) pogo block

Block A ... P: 16 x 16 channels

	a	b	c	d
1	GS2	P1	GND	PS1
2	PS2	GND	P2	GS1
3	GS4	P3	GND	PS3
4	PS4	GND	P4	GS3
5	GS6	P5	GND	PS5
6	PS6	GND	P6	GS5
7	GS8	P7	GND	PS7
8	PS8	GND	P8	GS7
9	GS10	P9	GND	PS9
10	PS10	GND	P10	GS9
11	GS12	P11	GND	PS11
12	PS12	GND	P12	GS11
13	GS14	P13	GND	PS13
14	PS14	GND	P14	GS13
15	GS16	P15	GND	PS15
16	PS16	GND	P16	GS15
17	GND	NC	GND	NC

Orientation mark •

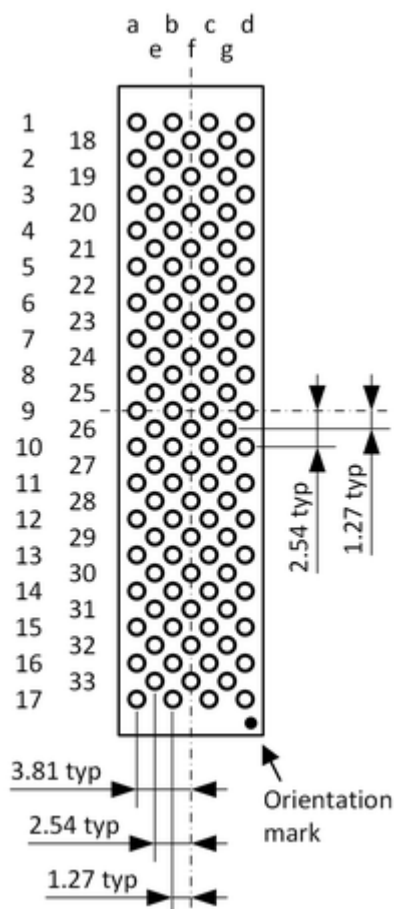
Layout and dimension



See also *Pogo pad and via design*.

EK751VHDA - Very High Density Pogo Cable for EK751 (8 x 32 channels)**Very-high-density (VHD) pogo cable assembly****Block A ... H: 8 x 32 channels**

	a	e	b	f	c	g	d
1	GSA2		PA1		GND		PSA1
18		PB1		PSB1		GSB1	
2	PSA2		GND		PA2		GSA1
19		GSB2		PB2		PSB2	
3	GSA4		PA3		GND		PSA3
20		PB3		PSB3		GSB3	
4	PSA4		GND		PA4		GSA3
21		GSB4		PB4		PSB4	
5	GSA6		PA5		GND		PSA5
22		PB5		PSB5		GSB5	
6	PSA6		GND		PA6		GSA5
23		GSB6		PB6		PSB6	
7	GSA8		PA7		GND		PSA7
24		PB7		PSB7		GSB7	
8	PSA8		GND		PA8		GSA7
25		GSB8		PB8		PSB8	
9	GSA10		PA9		GND		PSA9
26		PB9		PSB9		GSB9	
10	PSA10		GND		PA10		GSA9
27		GSB10		PB10		PSB10	
11	GSA12		PA11		GND		PSA11
28		PB11		PSB11		GSB11	
12	PSA12		GND		PA12		GSA11
29		GSB12		PB12		PSB12	
13	GSA14		PA13		GND		PSA13
30		PB13		PSB13		GSB13	
14	PSA14		GND		PA14		GSA13
31		GSB14		PB14		PSB14	
15	GSA16		PA15		GND		PSA15
32		PB15		PSB15		GSB15	
16	PSA16		GND		PA16		GSA15
33		GSB16		PB16		PSB16	
17	GND		NC		GND		NC

Orientation mark ***Layout and dimension**

See also XPS256 VHD pogo pad and via design and Pogo pad and via design.